HH-I-521F
September 4, 1980
SUPERSEDING
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July 24, 1972

FEDERAL SPECIFICATION

INSULATION BLANKETS, THERMAL (MINERAL FIBER, FOR AMBIENT TEMPERATURES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mineral fiber thermal insulation blankets for use in housing and other framed buildings at ambient temperatures. While products that comply with this specification may be used in various constructions, they are adaptable primarily, but not exclusively, to wood joist, rafters, and stud construction.

1.2 Classification.

1.2.1 Types and classes. Thermal insulation blankets shall be of the following types and classes as specified (see 6.2):

Type I - Blankets with no membrane coverings.
Type II - Blankets with a nonreflective vapor barrier membrane covering one principal face.

Class A - Membrane with flame spread of 25 or less.
Class B - Membrane with flame propagation resistance
- Critical radiant flux of 0.12 W/cm^2 or greater
Class C - Membrane not rated for flame propagation resistance; for use in non-exposed application only.

Type III - Blankets with a reflective vapor barrier membrane covering one principal face.

Class A - Membrane with flame spread of 25 or less.
Class B - Membrane with flame propagation resistance
- Critical radiant flux of 0.12 W/cm^2 or greater.
Class C - Membrane not rated for flame propagation resistance; for use in non-exposed applications only.

1.2.2 Sizes and thermal resistance (R) values. Sizes and thermal resistance (R) values of the blankets shall be in accordance with table I, as specified (see 6.2).

TABLE I. Sizes 1/ and thermal resistance (R) values

Type	R values 2/	Width (inches) excluding flanges	Length
I, II, III	11, 13, 19, 22 30, and 38	11, 15, 16, 19, 23, 24	Lengths of 100 inches or less in precut pieces. Lengths over 100 inches in rolls.

1/ All sizes listed are not available from all manufacturers. For other sizes and R values consult manufacturers.
2/ Insulation blankets of the same R value, but from different manufacturers, may vary appreciably in thickness (see 6.3). For R values other than those listed consult manufacturer. Thermal resistance (R) values apply to the mineral fiber only, as defined in 3.5.1.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-1065 - Barrier Material, Waterproofed, Flexible.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative biomonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, New York, Washington, DC; Philadelphia, Atlanta, Chicago, Kansas City, MO; Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C167 - Methods of Test for Thickness and Density of Blanket-or-Batt-Types Thermal Insulating Materials.
C177 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate.
C236 - Thermal Conductance and Transmittance of Built-Up Sections by Means of the Guarded Hot Box.
C518 - Standard Test Method for Steady-State Thermal Transmission by Means of the Heat Flow Meter.
C553 - Mineral Fiber Blanket and Felt Insulation (Industrial Type).
C653 - Determination of the Thermal Resistance of Low Density Mineral Fiber Blanket-Type Building Insulation.
C655 - Mineral Fiber Blanket Thermal Insulation for Wood Frame and Light Construction Buildings.
E84 - Method for Surface Burning Characteristics for Buildings Materials.
E96 - Methods of Test for Water Vapor Transmission of Materials in Sheet Form.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The basic material shall be fibers made from mineral substances such as rock, slag or glass processed from a molten state into an incombustible fibrous form.

3.2 Construction. The insulation blankets shall consist of flexible units composed of felted mineral fibers in rolls or flat cut pieces, with or without various combinations of attached membrane coverings, as specified.

3.2.1 Type I. Type I insulation blankets shall have no membrane coverings and shall have dimensions and physical properties to be self-supporting when placed between structural members having standard specaings. A partial covering for identification purposes shall be permissible.

3.2.2 Types II and III. Types II and III insulation blankets shall have a nonreflective or reflective membrane covering, as applicable, on one principal face. The membrane coverings shall, unless otherwise specified, be vapor resistant and shall extend beyond the longitudinal edges of the insulation to provide flanges for attachment to structural members. A vapor permeable membrane covering on the opposite face may be provided at the supplier's option.

3.3 Membrane coverings.

3.3.1 Vapor permeance. When tested in accordance with 4.4.1, vapor resistant membrane coverings shall have a vapor permeance of not more than 1 perm and vapor permeable membrane coverings shall have a vapor permeance of not less than 5 perms.

3.4 Dimensional data.

3.4.1 Thickness. The thickness of insulation shall be not less than that required to provide the thermal resistance (R) value specified (see table I, 3.5.1 and 4.4.9.) When thickness limitations are imposed by the size of structural members, the maximum acceptable thickness of insulation to provide the required thermal resistance (R) value shall be as specified (see 6.2).

3.4.2 Dimensional tolerance.

3.4.2.1 Length. The tolerance on the specified length shall be as follows:

100 inches or less	-1/2 inch
over 100 inches	-0.5 percent

3.4.2.2 Width. The tolerance on the specified width, excluding any attaching flanges, shall be as follows:

Type I	$\pm \frac{1/2 \text{ inch}}{0}$
Type II and III	$\pm$ 1/2 inch

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3.5 Physical requirements.

3.5.1 Thermal resistance. When determined in accordance with 4.4.9 the thermal resistance (R) value of blankets, excluding membrane coverings, expressed as a reciprocal of thermal conductance (C) shall average not more than 5 percent below the R value specified for a minimum of four specimens tested, (see 1.2.2 and 4.4.9).

3.5.2 Flame spread index. Class A membrane coverings on Type II and Type III insulation blankets shall have a flame spread not greater than 25 when tested in accordance with 4.4.6.

3.5.3 Critical radiant flux. Insulation blankets shall have a critical radiant flux-flame propagation resistance equal to or greater than 0.12 W/cm^2 when tested in accordance with 4.4.7. Blankets with Class B membrane coverings shall be tested twice, once with the vapor barrier membrane exposed, and once with the opposite principal face exposed.

3.5.3.1 Caution statement. All Class C membrane coverings shall be marked at intervals not exceeding 4 linear feet for 48 inch lengths and at intervals not exceeding 8 linear feet for lengths of 96 inches or greater, with a caution statement that the membrane facing is combustible and must not be left exposed.

3.5.4 Smoldering combustion. When tested in accordance with 4.4.8, the insulation shall show no evidence of flaming combustion and the weight loss shall be not greater than 15 percent.

3.5.5 Moisture adsorption. When tested in accordance with 4.4.2, the moisture adsorption of the insulation shall be not more than 5 percent by weight.

3.5.6 Odor emission. An odor classified as objectionable and as either strong or very strong by more than two members of the panel when tested in accordance with 4.4.3, shall constitute unacceptable odor emission and failure of the test.

3.5.7 Corrosiveness. When tested in accordance with 4.4.4, the steel test plate in contact with the insulation shall show no greater corrosion than the comparative steel plate in contact with sterile cotton which has been tested in the same manner.

3.5.8 Fungi resistance. The insulation shall have growth no greater than that observed on the comparative items when tested in accordance with 4.4.5.

3.6 Identification marking.

3.6.1 Insulation blankets. Insulation blankets shall be marked with the manufacturer's name or trademark, the thermal resistance (R) value of the insulation, and the thickness to obtain this thermal resistance (R) value. On each type I insulation blanket, the thermal resistance (R-value) shall be code-branded, be printed on a partial facing, or indelibly marked. On types II and III insulation blankets, the identification marking shall be printed on the vapor resistant membrane covering at intervals not exceeding 8 linear feet.

3.6.2 Shipping bags and wrapping. Shipping bags and wrapping (see 5.1) for insulation blankets shall be marked with the manufacturer's name or trademark, the length, width, thickness, the R value of the insulation, and the square feet of insulation.

3.6.3 Shipping containers. Shipping containers (see 5.2) for insulation blankets shall be marked with the manufacturer's name or trademark, the length, width, and the R value of the insulation.

3.7 Workmanship. Insulation shall be clean, dry, and free of extraneous materials. Fibers shall be evenly distributed and insulation and facings shall be sufficiently coherent to be unaffected by handling and installation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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4.2 Sampling. Each unit of product selected in accordance with 4.2.1 shall be a total of 8 linear feet of insulation.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with level S-2 of MIL-STD-105.

4.2.2 Sampling for tests. Sampling for tests shall be in accordance with level S-1 of MIL-STD-105.

4.3 Examination.

4.3.1 End item. Samples selected in accordance with 4.2.1 shall be examined for defects listed in table II. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Type of insulation not as specified	X	
Dimensions not as specified or not within tolerances	X	
Unclean or wet insulation	X	
Foreign material in insulation	X	
Identification marking not as required (see 3.6)	X	
Facing material missing, badly torn, punctured or inadequately bonded to insulation	X	
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.3.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.4 Test methods.

4.4.1 Vapor permeance of membranes. Vapor permeance shall be determined in accordance with procedure A of ASTM E 96 (see 3.3.1).

4.4.2 Moisture adsorption. Moisture adsorption shall be determined in accordance with ASTM C 553, section 15 (see 3.5.5).

4.4.3 Odor emission. Odor emission shall be determined as follows. Subject a minimum of 50 g of the insulation to a soaking temperature of 150° F (65° C) for 30 minutes in a closed container. The container shall be made of stainless steel or glass jars with all metallic lids. Take care to use containers that will not produce odor. A panel 1/ five persons with normal odor perception shall examine for odor using the following questions:

1. Was the odor perceptible?
2. Was the odor objectionable, pleasant or otherwise?
3. Was the odor weak, strong or very strong?

1/ Panel members shall be persons who have demonstrated an ability to detect odors accurately and consistently.

4.4.4 Corrosiveness. Determination of corrosion shall be in accordance with the following test procedure (see 3.5.7):

Corrosiveness - Test specimen--Two specimens shall be used, each measuring 1-1/2 inches by 4-1/2 inches by approximately 1/2 inch thick when compressed against the metal test plates under the conditions of this test. Apparatus--Two steel test plates 1 inch wide, 4 inches long and 0.020 inch thick. They shall be bright No. 2 finish, cold rolled strip steel, quarter hard, temper No. 3. Four pieces, 1-1/2 inches by 4-1/2 inches, Type 304 stainless steel woven wire screen, 7/16" open square grid, 1/16 inch wire. Four No. 12 rubber bands. Clean humidity test chamber.

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Procedure

The steel test plates shall be cleaned until completely free of water breaks by a five minutes vapor degreasing followed by a 15 minute hot caustic bath (15 percent KOH solution), rinsed thoroughly in distilled water and immediately dried. One piece of cold rolled steel shall be placed between two insulation specimens. This assembly is then compressed between two pieces of woven wire cloth and secured at each end with a No. 12 rubber band. The compressed thickness of this assembly shall measure 1 inch ± 1/8 inch. The other piece of cold rolled steel is placed between two 1-1/2 inch by 4-1/2 inch pieces of sterile cotton which have been solvent extracted in ACS reagent grade acetone for 48 hours, and vacuum dried at low heat. The cotton is secured against the metal test plate in the same manner as the insulation specimen. The test specimen and extracted sterile cotton covered plate shall be suspended vertically in an atmosphere free of contaminants having a relative humidity of 95 percent ± 3 percent, and a temperature of 120° ± 3° F, for 7 days, and then examined for corrosion.

4.4.5 Fungi resistance. Fungi resistance shall be determined in accordance with the following test procedure:

APPARATUS

The apparatus required to conduct this test consists of chambers of cabinets together with auxiliary instrumentation capable of maintaining the specified conditions of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the test item. There shall be free circulation of air around the test item, and the surface area of fixtures supporting the test item shall be kept to a minimum.

PROCEDURES

Preparation of mineral-salts solution. The solution shall contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7g
Potassium monohydrogen orthophosphate (K_2HPO_4)	0.7g
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0.7g
Ammonium nitrate (NH_4NO_3)	1.0g
Sodium chloride (NaCl)	0.005g
Ferrous sulfate ($FeSO_4 \cdot 7H_2O$)	0.002g
Zinc sulfate ($ZnSO_4 \cdot 7H_2O$)	0.002g
Manganous sulfate ($MnSO_4 \cdot 7H_2O$)	0.001g
Distilled water	1000ml

Sterilize the mineral salts solution by autoclaving at 121° C (250° F) for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal solution of NaOH so that after sterilization the pH is between 6.0 and 6.5. Prepare sufficient salts solution for the required tests.

Reagent grade chemicals shall be used in all tests. Unless otherwise specified, it is intended that all reagents shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.

Unless otherwise specified, references to water shall be understood to mean distilled water or water of equal purity.

Preparation of mixed spore suspension. The following test fungi shall be used:

Fungi	ATTC 1/	NLABS 2/
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funiculosum	11797	474
Chaetomium globosum	6205	459

1/ American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852.
2/ Pioneering Research Division, U.S. Army Natick Laboratories, Natick, Massachusetts 01760.

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Maintain cultures of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of Chaetomium globosum shall be cultured in strips of filter paper on the surface of mineral salts agar. (Mineral salts agar is identical to the mineral salts solution described above, but contains in addition 15.0g of agar per liter.) The stock culture may be kept for not more than 4 months at 6° ± 4° C (43° ± 7° F) at which time subcultures shall be made, and new stocks shall be selected from the subcultures. If genetic or physiological changes occur, obtain new cultures as specified above. Subcultures used for preparing new stock cultures or the spore suspension shall be incubated at 30° C (86° F) for 9 days or longer. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a 10 ml portion of a sterile solution containing 0.05g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate or sodium lauryl sulfate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125 ml glass-stoppered Erlenmeyer flask containing 45 ml of sterile water and 50 to 75 solid glass beads, 5 mm in diameter. Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps. Filter the dispersed fungal spore suspension through a 6 mm layer of glass wool contained in a glass funnel, into a sterile flask. This process is intended to remove large mycelial fragments and clumps of agar which could interfere with the spraying process. Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid. Resuspend the residue in 50 ml of sterile water and centrifuge. Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral-salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ± 200,000 spores per ml as determined with a counting chamber. Repeat the operation for each organism used in the test and blend equal volumes of the resultant spore suspensions to obtain the final mixed spore suspension. The spore suspension may be prepared fresh each day or may be held at 6° ± 4° C (43° ± 7° F) for not more than 7 days.

Viability of inoculum control. With each daily group of tests, place each of three pieces of sterilized filter paper, 1-inch square, on hardened mineral salts agar in separate Petri dishes. Inoculate these with the spore suspension by spraying the suspension from a sterilized atomizer 1/ until initiation of droplet coalesence. Incubate these at 30° C (86° F) at a relative humidity of not less than 85 percent and examine them after 7 days of such incubation. There should be copious growth on all three of the filter paper control specimens. Absence of such growth requires repetition of the test.

1/ An atomizer capable of providing 15,000 ± 3,000 spores per square centimeter.

Control items. In addition to the viability of inoculum control, known susceptible substrates shall be inoculated along with the test item to insure that proper conditions are present in the incubation chamber to promote fungus growth. The control items shall consist of cotton duck 8.25-ounce strips that are 1.25 inches wide; that have been dipped into a solution containing 10 percent glycerol, 0.1 percent potassium dihydrogen orthophosphate (KH_2PO_4), 0.1 percent ammonium nitrate (NH_4NO_3), 0.025 percent magnesium sulfate ($MgSO_4 \cdot 7H_2O$), and 0.05 percent yeast extract (pH 5.3); and from which the excess liquid has been removed. The strips should be hung to air dry before being inoculated and placed into the chamber.

Comparative items. Sections of southern pine approximately 3 inches wide by 6 inches long shall be employed as comparative items to determine the relative rate of growth on items being tested.

Inoculation of test, comparative, and control items. Mount the test, comparative, and control items on suitable fixtures or suspend them from hangers as is applicable. Precondition the chamber and its contents at 29° C (84° F) and 95 percent relative humidity for at least 4 hours. Inoculate the test, comparative, and control items with the mixed fungus spore suspension by spraying it on and into these items in the form of a fine mist from a previously sterilized atomizer or nebulizer. In spraying the test, comparative, and control items, care should be taken to spray all surfaces. (For loose-fill materials, thorough spraying of the top surface will suffice.) If surfaces are nonwetting, spray until initiation of droplet coalesence. Incubation is to be started immediately following the inoculation.

Incubation. Maintain the test chamber at 29° C (84° F) and at a relative humidity of 95 percent minimum during the life of the test. Keep the test chamber closed during the incubation period except during inspection. After 14 days, inspect the control items. If control items do not show an abundant growth at this time, the entire test shall be repeated. If sufficient growth is present on the control items, continue the test for a period of 28 days from the time of incubation.

Inspection. At the end of the incubation period, the test and comparative items shall be removed from the test chamber and examined at 40X magnification. Test items which have growth greater than that on the comparative items shall be considered to have failed. Test items on which the growth is not greater

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than that on the comparative items shall be considered to have passed.

4.4.6 Flame spread index. Class A membrane coverings shall be tested for flame spread index in accordance with ASTM E84 (see 3.5.2).

4.4.7 Critical radiant flux. Critical radiant flux shall be determined in accordance with the following test procedure: Standard Method of Test for Critical Radiant Flux of Exposed Attic Floor Insulation using a radiant heat energy source.

APPARATUS

The basic elements of the test chamber are: (1) an air-gas fueled radiant heat energy panel inclined at 30° and directed at (2) a horizontally-mounted attic floor insulation specimen. The radiant panel generates a radiant energy flux distribution ranging along the 100-cm length of the test specimen from a nominal maximum of 1.0 watts/cm^2 to a minimum of 0.1 watts/cm^2. The test is initiated by open flame ingnition from a pilot burner. The distance burned to flame-out is converted to watts/cm^2 from the flux profile graph (Figure 6) and reported as critical radiant flux, watts/cm^2.

The Radiant Panel Test Chamber shall be Constructed and Instrumented as follows:

The radiant panel test chamber employed for this test shall be located in a draft protected area maintained at 21 ± 3° C (69.8 ± 5.4° F) and relative humidity of 50 ± 20 percent. The radiant panel test chamber (Figure 1 and 2) shall consist of an enclosure 140 cm (55 in.) long by 50 cm (19 1/2 in.) deep by 71 cm (28 in.) above the test specimen. The sides, ends, and top shall be of 1.3 cm (1/2 in.) nominal calcium silicate board such as Marinite I, 0.74 g/cm^3 (46 lb/ft^3) nominal density, with a thermal conductivity at 177° C (350° F) of 1.11 cal (g)/hr. cm 20 C/cm [0.89 Btu/(hr)(ft^2)(°F/in)]. One side shall be provided with an approximately 10 cm X 110 cm (4 x 44 inches) draft tight fire resistance glass window so that the entire length of the test specimen may be observed from outside the fire test chamber. On the same side and below the observation window is a door which, when open, allows the specimen platform to be moved out for mounting or removal of test specimens. A draft tight fire resistant observation window may be installed at the low flux end of the chamber.

The bottom of the test chamber shall consist of a sliding steel platform which has provisions for rigidly securing the test specimen holder in a fixed and level position. The free, or air access, area around the platform shall be in the range of 1950-3550 cm^2 (300-500 square inches). The top of the chamber shall have an exhaust stack with interior dimensions of 12.5 cm (5 inches) wide by 38 cm (15 inches) deep by 30 cm (12 inches) high at the opposite end of the chamber from the radiant energy source. The radiant heat energy source shall be a panel of porous refractory material mounted in a cast iron frame, with a radiation surface of 30.5 x 45.7 cm (12 by 18 inches). It shall be capable of operating at temperatures up to 816° C (1500° F). The panel fuel system shall consist of a venturi-type aspirator for mixing gas and air at approximately atmospheric pressure, a clean dry air supply capable of providing 28.3 NTP m^3 per hr (1000 Standard Cubic Feet per Hour) at 7.6 cm (3.0 inches) of water, and suitable instrumentation for monitoring and controlling the flow of fuel to the panel.

The radiant heat energy panel shall be mounted in the chamber at 30° to the horizontal specimen panel. The horizontal distance from the 0 mark on the specimen fixture to the bottom edge (projected) of the radiating surface of the panel is 8.9 cm (3-1/2 inches). The panel to speciment vertical distance is 14 cm (5-1/2 inches) (see Figure 1). The angle and dimensions given above are critical in order to obtain the required radiant flux. The radiation pyrometer for standardising the thermal output of the panel shall be suitable for viewing a circular area 25.4 cm (10 inches) in diameter at a range of about 1.37 m (54 inches). It shall be calibrated over the black body temperature range of 490-510° C (914-950° F) in accordance with the procedure described in Appendix A. A high impedance potentiometer voltmeter with a suitable millivolt range shall be used to monitor the output of the radiation pyrometer described.

The dummy holder (see Figure 3), shall be constructed from heat-resistance stainless steel (ANSI Type 300 (UNA-N08330) or equivalent, thickness 0.198 cm (0.078 inches), having overall dimension of 115 cm (45 inches) by 32 cm (12-3/4 inches) with a specimen opening of 20 cm (7.9 inches) by 100 cm (40 inches). Six slots are cut in the flange on either side of the holder to reduce warping. The holder is fastened to the platform with two stud bolts at each end.

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The pilot burner, used to ignite the specimen shall be a propane venturi torch with an axially symmetric burner tip having a propane supply tube with an orifice diameter of 0.0076 cm (0.003 inches). In operation, the propane flow is adjusted to give a pencil flame blue inner cone length of 1.3 cm (1/2 inch). The pilot burner is positioned so that the flame generated will impinge on the centerline of the specimen at the 0 distance burned point at right angles to the specimen length (see Figures 1 and 2). The burner shall be capable to being swung out of the ignition position so that the flame is horizontal and at least 5 cm (2 inches) above the specimen plane.

Two 3.2 cm (1/8 inch) stainless steel sheathed grounded junction chromel alumel thermocouples are located in the Flooring Radiant Panel Test Chamber (see Figures 1 and 2). Thermocouples shall be kept clean to insure accuracy of readout. The chamber thermocouple is located in the longitudinal central vertical plane of the chamber 2.5 cm (1 inch) down from the top and 10 cm (4 inches) back from the inside of the exhaust stack. The exhaust stack thermocouple is centrally located 15.2 cm (6 inches) from the top. An indicating potentiometer with range of 100-500° C (212-932° F) may be used to determine the chamber temperatures prior to a test.

An exhaust duct with a capacity of 28.3-85 NTP m^3 per minute (1000-3000 SCFM) decoupled from the chamber stack by at least 7.6 cm (3 inches) on all sides and with an effective area of the canopy slightly larger than the plane area of the chamber with the specimen platform in the out position shall be used to remove combustion products from the chamber. With the panel turned on and dummy speciment in place, there shall be no measurable difference in air flow through the chamber stack with the exhaust on or off.

The dummy specimen which is used in the flux profile determination shall be made of 1.9 cm (3/4 inch) inorganic 0.58 g/mcm^3 (46 lbs/ft^3) nominal density calcium silicate board, such as Marinite I (see Figure 3). It is 25 cm (10 inches) wide by 107 cm (42 inches) long with 2.7 cm (1-1/16 inches) diameter holes centered on and along the centerline at the 10, 20, 30, 40, 50, 60, 70, 80, 90 cm locations, measured from the maximum flux end of the specimen. The total heat flux transducer used to determine the flux profile of the chamber in conjunction with the dummy specimen should be of the Schmidt-Boelter type, have a range of 0-1/5 watts/cm^2 (0-1/32 Btu/ft^2 s), and shall be calibrated over the operating flux level range of 0.10 to 1.5 watts/cm^2 in accordance with the procedure outlined in Appendix A. A source of 15-25° C (59-77° F) cooling water shall be provided for this instrument. A high impedance or potentiometric voltmeter with a range of 0-10 m.v. and reading to 0.01 m.v. shall be used to measure the output of the total heat flux transducer during the flux profile determination. A timer shall be used for measuring preheat and pilot contact time.

SAFETY PROCEDURES

The possibility of a gas-air fuel explosion in the test chamber should be recognized. Suitable safeguards consistent with sound engineering practice should be installed in the panel fuel supply system. These may include one or more of the following (1) a gas feed cut-off activated when the air supply fails, (2) a fire sensor directed at the panel surface that stops fuel flow when the panel flame goes out, (3) a commercial gas water heater or gas-fired furnance pilot burner control thermostatic shut-off which is activated when the gas supply fails or other suitable and approved device. Manual reset is a requirement of any safeguard system used. In view of the potential hazard from products of combustion, the exhaust system must be so designed and operated that the laboratory environment is protect from smoke and gas. The operator should be instructed to minimize his exposure to combustion products by following sound safety practice, e.g., insure exhaust system is working properly, wear appropriate clothing including gloves, et al.

TEST SPECIMENS

A test specimen shall consist of a section of blanket cut so as to fill the specimen tray. The surface of the insulation subjected to the test shall represent, insofar as possible, the surface that would be exposed in actual installation practice. Materials which incorporate breather paper, that is intended to be exposed after installation, shall be tested with the breather paper in place and exposed. When Class B blankets are to be tested, two test specimens shall be prepared. One specimen shall have the membrane side exposed, and the other shall have the side opposite the membrane exposed. Both specimens shall be subjected to the test.

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Radiant Heat Energy Flux Profile Standardization

Mount the dummy specimen in the mounting frame and attach the assembly to the sliding platform. With the sliding platform out of the chamber, ignite the radiant panel. Allow the unit to heat for 1 hour. The pilot burner is off during this determination. Adjust the fuel mixture to give an air-rich flame. Make fuel flow settings to bring the panel to an apparent black body temperature as measured by the radiation pyrometer of about 500° C (932° F), and the chamber temperature to about 180° C (356° F). When equilibrium has been established, move the specimen platform into the chamber. Allow 0.5 hours for the closed chamber to equilibrate.

Measure the radiant heat energy flux level at the 40 cm point with the total flux meter instrumentation. This is done by inserting the flux meter in the opening so that its detecting plane is 0.16-0.32 cm (1/16-1/8 inch) above and parallel to the plane of the dummy specimen and reading its output after 30 ± 10 seconds. If the level is within the limits specified, the flux profile determination is started. If it is not, make the necessary adjustments in panel fuel flow. A suggested flux profile data log format is shown in Figure 5.

The test shall be run under chamber operating conditions which give a flux profile as shown in figure 6. The radiant heat energy incident on the dummy specimen shall be between 0.87 and 0.95 W/cm^2 (0.77 and 0.83 Btu/fts sec) at the 20 cm point, between 0.48 and 0.52 W/cm^2 (0.42 and 0.46 Btu/ft^2 sec) at the 40 cm point and between 0.22 and 0.26 W/cm^2 (0.19 and 0.23 Btu/ft^2 sec) at the 60 cm point. Insert the flux meter in the 10 cm opening following the procedure given above. Read the 10 seconds and proceed to the 20 cm point. Repeat the 10 cm procedure. The 30 - 90 cm flux levels are determined in the same manner. Following the 90 cm measurement, make a check reading at 40 cm. If this is within the limits set forth, the test chamber is in calibration and the profile determination is completed. If not, carefully adjust fuel flow, allow 0.5 hours for equilibrium and repeat the procedure. Plot the radiant heat energy flux data as a function of distance along the specimen plane on rectangular coordinate graph paper. Carefully draw the best smooth curve through the data points. This curve will hereafter be referred to as the flux profile curve.

Determine the open chamber apparent black body and chamber temperatures that are identified with the standard flux profile by opening the door and moving the specimen platform out. Allow 0.5 hours for the chamber to equilibrate. Read the radiation pyrometer output and record the apparent black body temperature in °C. This is the temperature setting that can be used in subsequent test work in lieu of measuring the radiant flux at 20 cm, 40 cm, and 60 cm using the dummy specimen. The chamber temperature also shall be determined again after 0.5 hours and is an added check on operating conditions.

CONDITIONING

Test specimens shall be conditioned to equilibrium or a minimum of 48 hours, whichever is greater, at 21 ± 3° C (69.8 ± 5.4° F) and a relative humidity of 50 ± 5 percent immediately prior to testing.

TEST PROCEDURES

With the sliding platform out of the chamber, ignite the radiant panel. Allow the unit to heat for 1 hour. It is recommended that a sheet of inorganic millboard be used to cover the opening when the hinged portion of the front panel is open and the specimen platform is moved out of the chamber. The millboard is used to prevent heating of the specimen and to protect the operator. Read the panel apparent black body temperature and the chamber temperature. If these temperatures are in agreement to within ± 5° C with those determined previously, as above, the chamber is ready for use.

Mount the specimen tray with insulation on the sliding platform and position with stud bolts (see Figure 7). Ignite the pilot burner, move the specimen into the chamber, and close the door. Start the timer. After 2 minutes preheat, with the pilot burner on and set so that the flame is horizontal and 5 cm above the specimen, bring the pilot burner flame into contact with the center of the specimen at the 0 mark. Leave the pilot burner flame in contact with the specimen for 5 minutes, then remove to a position 5 cm above the specimen and leave burning until the test is terminated.

If the specimen does not ignite within 5 minutes following pilot burner flame application, the test is terminated by extinguishing the pilot burner flame. For specimens that do ignite, the test is continued until the flame goes out. When the test is completed, the door is opened, the specimen platform is pulled out.

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Measure the distance burned, i.e., the point of farthest advance of the flame front, to the nearest 0.1 cm. From the flux profile curve, convert the distance to watts/cm^2 critical radiant heat flux at flame out. Read to two significant figures. A suggested date log format is shown in Figure 8. Remove the specimens tray from the moveable platform. The succeding test can begin as soon as the panel apparent black body temperature and chamber temperature are verified. The specimen tray should be at room temperature prior to insertion of the next specimen.

4.4.8 Smoldering combustion. Smoldering combustion test shall be determined in accordance with the following test procedure: Standard Method of Test for Smoldering Combustion Characteristics of Materials Used for Thermal Insulation.

APPARATUS

The specimen holder shall be an open-top 20 ± 0.2 cm square box, 10 ± 0.2 cm in height, fabricated from a single piece of 0.61 ± 0.08 mm thick (24 U.S. Standard gauge) stainless steel sheet with the vertical edges of the box overlapped, not to exceed to 7 mm in seam width, and soldered so as to be watertight. The specimen holder during test use shall rest upon a pad of glass fiberboard having dimensions equal to or greater than those of the bottom of the specimen holder. The glass fiberboard shall be approximately 2.5 cm thick with adensity of 40 ± 8 kg/m^3 (Note 1). Note 1: Glass fiberboard which meets Federal Specification HH-I-558B (Form A, class 1, plain faced) is suitable for this purpose.

Ignition source. The ignition source shall be a cigarette without filter tip made from natural tobacco, 85 ± 2 mm long with a tobacco packing density of 0.270 ± 0.020 g/cm^3 and a total weight of 1.1 ± 0.1 gm.

Balance. A balance of 1 kg capacity, accurate at least to 0.1 g, and provided with a suitable scale plan is required.

Test area. The test area shall be draft-protected and equipped with a suitable system for exhausting smoke and/or noxious gases produced by testing. Air velocities as measured by a hot wire anemometer in the vincinity of the surface of the sample shall not exceed 0.5 m/sec. The test area shall be maintained at 23 ± 5° C and 50 ± 5 percent relative humidity.

TEST PROCEDURE

Samples and cigarettes shall be conditioned in air at a temperature of 21 ± 3° C (69.8 ± 5.4° F) and a relative humidity of 50 ± 5 percent for at least 12 hours prior to test. Samples and cigarettes shall be removed from any packaging and exposed in a suitable manner to permit free movement of air aound them during conditioning. A section shall be cut from sample blankets which fits into the test holder without folding. The cut section shall be placed in the holder, and the holder with sample shall be weighed to the nearest 0.2 g or less. The holder shall then be placed on the insulated pad. Insert a pointed rod of 8 mm diameter vertically into the appropriate center of the material and withdraw it, forming an appropriate cavity for the ignition source. A well lit cigarette, burned not more than 8 mm (0.3 inch), shall be inserted using tweesers into the formed cavity, with the lit end upward and flush with the sample surface. Burning of the cigarette and specimen shall be allowed to proceed undisturbed in the test area for at least 2 hours or until the smoldering is no longer progressing.

After completion of burning and after the holder has cooled down to 25° C, the specimen holder with its material residue shall be weighed, at least to the nearest 0.2 g, and the percent weight loss of the original specimen calculated, The weight of the cigarette residue is ignored in this calculation.

4.4.9 Thermal resistance. Thermal resistance shall be calculated, using ASTM C 653, from the thermal conductivity or the thermal conductance as determined in accordance with ASTM C 518. The mean temperature shall be 75° F (23.9° C) and the temperature difference shall be a minimum of 40° F (see 3.5.1).

4.4.9.1 Calibration. The C518 apparatus shall be calibrated using reference materials available from the National Bureau of Standards.

4.5 Certification. A certificate of compliance, signed by a responsible official of the contractor, shall be acceptable as proof that the product being offered meets the requirements of section 3.5 provided the contractor shall have available for inspection upon request actual test results, acceptable to the contract-

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ing officer, which indicate that tests have been performed to substantiate the certification. Supportive test data shall be not more than 3 years old at the date of the invitation for bids. The certificate shall state that the tests on which the data is furnished have been performed on products manufactured from the same basic materials and using the same manufacturing processes as the items being offered. It shall also state that any changes in materials or processes that affect the performance of the product will be reported to the contracting officer. The Government reserves the right to require additional testing and certification by the contractor when such changes are made or when otherwise deemed necessary.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Precut pieces. Precut pieces of insulation furnished in widths, lengths, and quantities as specified (see 6.2), shall be packaged in a bag or wrapped with material meeting the requirements of PPP-B-1055, B-1 or B-2. The wrap or bag shall be secured by stitching, tape, adhesive, or staples.

5.1.1.2 Rolls. Rolls of insulation furnished in widths, lengths, and quantities as specified (see 6.2), shall be packaged in a bag or wrapped with material meeting the requirements of PPP-B-1055, B-1, or B-2. The wrap or bag shall be secured by stitching, tape, adhesive, or staples.

5.1.2 Commercial. The blanket insulation shall be packaged in accordance with normal commercial practice. The material shall be packaged in bags or wrapped to insure safe delivery at destination, to provide a safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. When specified (see 6.2), insulation of like description shall be packed in a close-fitting box conforming to PPP-B-636, V3C or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall box shall not exceed 200 pounds. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. When specified (see 6.2) insulation of like description packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, style optional. The boxes shall be closed and reinforced in accordance with the appendix to PPP-B-636.

5.2.3 Commercial. The blanket insulation shall be packed in accordance with normal commercial practice. The material shall be packed in containers to insure safe delivery at destination, to provide a safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. All types of insulation are intended for use as thermal insulation in housing and other buildings at ambient temperatures. Insulation with class A membrane is intended for all applications where a flame spread not greater than 25 is required. Insulation with class B membrane may be utilized on exposed surfaces in normal attic applications.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Length, width, and R value required (see 1.2.2 and table 1).
(d) When required, the maximum acceptable thickness to provide the specified R value (see 3.4.1).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) Quantity of insulation required (see 5.1.1 and 5.1.2).
(g) When boxes are required for shipping containers (see 5.2.1 and 5.2.2).
(h) Marking required (see 5.3).

6.3 Insulation thickness. Insulation is produced by different manufacturers at different densities, therefore, there is a wide variation in thickness for the same R value between various manufacturers. No effort should be made to relate any specific R value to any specific thickness.

MILITARY CUSTODIANS:

Navy - YD
Air Force - 84

Review activities:

Navy - YD
Air Force - 84

User activities:

Army - ME, CE
Navy - None
Air Force - None

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA - FSS

Civil Agency Coordinating Activities:

COMMERCE - NBS
GSA - FSS, PCD
HUD - TCS
INTERIOR - BPA
DOE

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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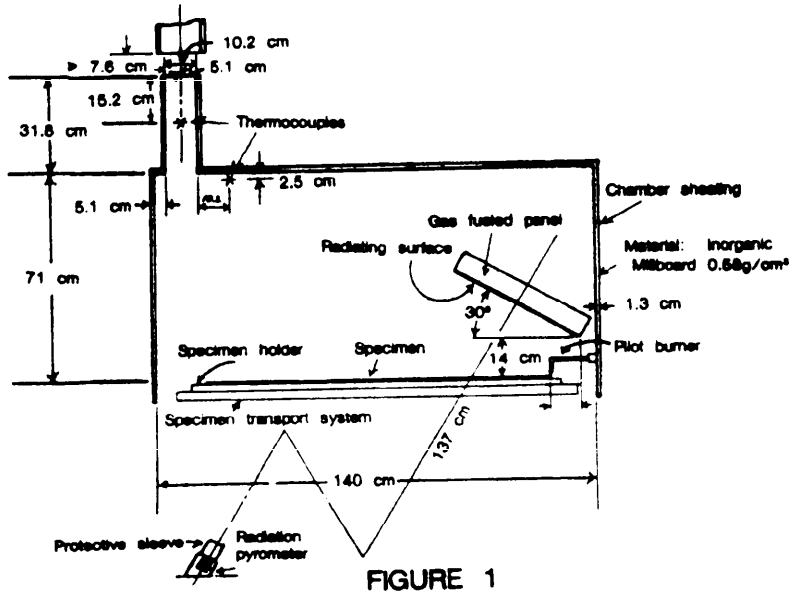
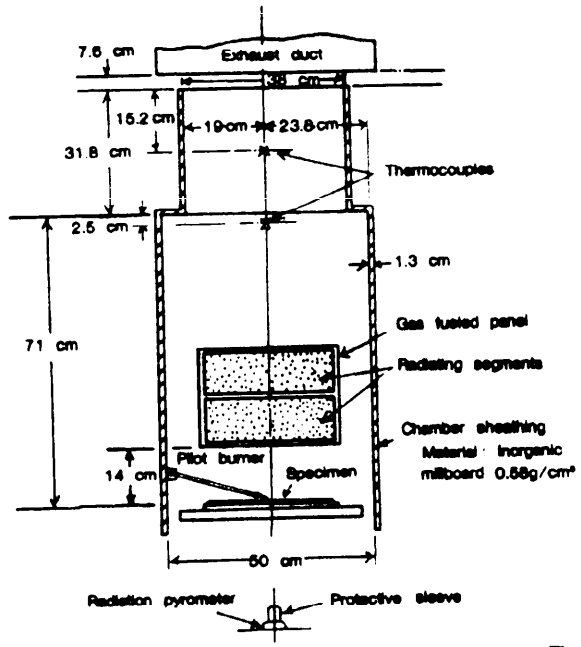


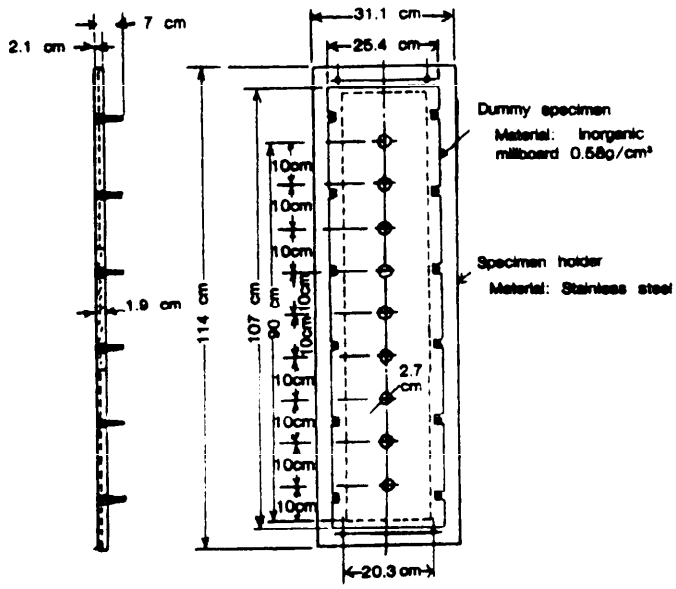
FIGURE 1



Flooring Radiant Panel Tester Schematic Low Flux End Elevation

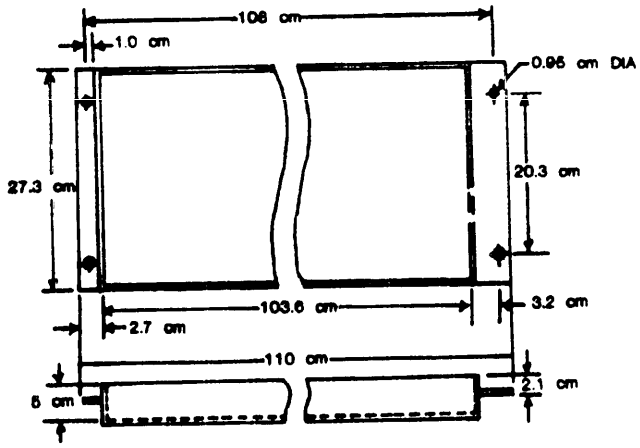
FIGURE 2

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Dummy Specimen in Specimen Holder

FIGURE 3



Specimen tray

FIGURE 4

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RADIANT FLUX PROFILE

Date _____

Black Body Temperature _____ m.v. _____ °C (°F)

Gas Flow _____ NTPm³H (SCFH) Air Flow _____ NTPm³H (SCFH)

Room Temperature _____ °C (°F)

Air Pressure _____ Gas _____ cm (in) of H_2O

Flux Meter

Radiometer No. _____ Conversion Factor _____

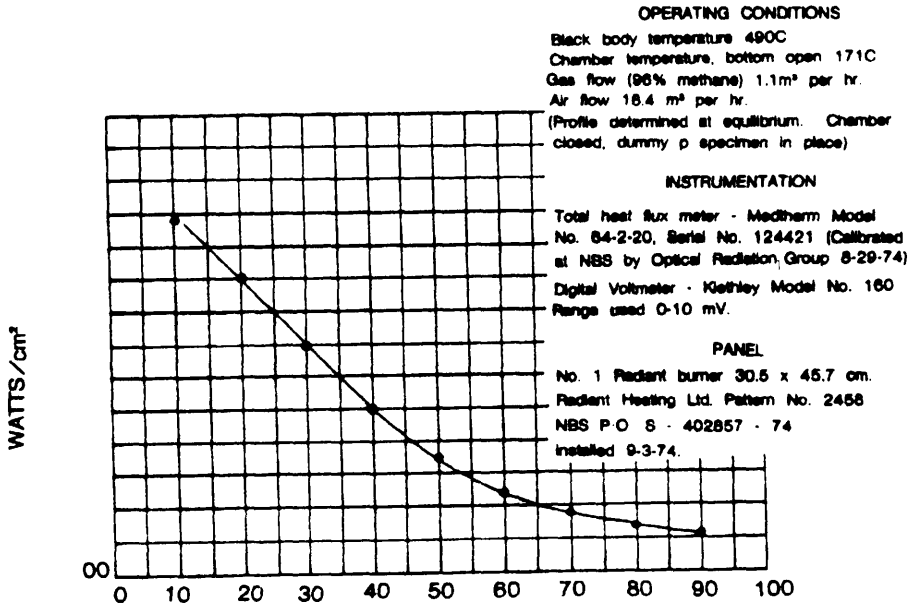
from Calibration on _____

Distance (cm)	MV	Watts/cm²
10		
20		
30		
40		
50		
60		
70		
80		
90		

Signed _____

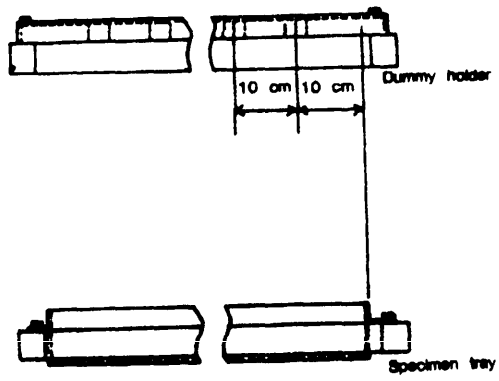
FIGURE 5. Flux Profile Data Log Format

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Standard Radiant Heat Energy Flux Profile

FIGURE 6



Specimen Tray Mounting Position

FIGURE 7

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Test Number _____ Date _____ Time _____

Laboratory _____

Specimen Identification/Code No. _____

Test Assembly _____

Panel: Angle _____ ° Temperature _____ °C (°F)

Flow: Gas _____ $NTPm^3H$ (SCFH) Air _____ $NTPm^3H$

Pressure,cm (in) H_2O: Initial, Air _____ Gas _____

Chamber Temperature (Initial) _____ °C (°F)

Room: Temperature _____ °C (°F) Hood Draft _____ cm (in) water

Flame Front Out _____ min.

All Flame Out _____ min.

Total Burn Length _____ cm (in)

Critical Radiant Flux watts/cm^2 _____

Flux Profile Reference _____

Observations:

Signed _____

FIGURE 8. Insulation Radiant Panel Test Data Log Format

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APPENDIX A

PROCEDURE FOR CALIBRATION OF RADIATION INSTRUMENTATION

Radiation pyrometer

Calibrate the radiation pyrometer by means of a conventional black body enclosure placed within a furnace and maintained at uniform temperatures of 490, 50, and 510° C (914, 932, 950° F). The black body enclosure may consist of a closed chromel metal cylinder with a small sight hole in one end. Sight the radiation pyrometer upon the opposite end of the cylinder where a thermocouple indicates the black body temperature. Place the thermocouple within a drilled hole and in good thermal contact with the black body. When the black body enclosure has reached the appropriate temperature equilibrium, read the output of the radiation pyrometer. Repeat for each temperature.

Total heat flux meter

The total flux meter shall be calibrated by the National Bureau of Standards, (Direct request for such calibration services to the: Optical Radiation Section, 232.04, National Bureau of Standards, Washington, DC 20234), or alternatively, its calibration shall be developed by transfer calibration methods with an NBS calibrated flux meter. This latter calibration shall make use of the radiant panel tester as the heat source. Measurements shall be made at each of the nine dummy specimen positions and the mean value of these results shall constitute the final calibration.

It is recommended that each laboratory maintain a dedicated calibrated reference flux meter against which one or more working flux meters can be compared as needed. The working flux meters should be calibrated according to the procedure of A2.1 at least once per year.

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